

3RD

SEMESTER

B. Tech. (Computer Science and Engineering)
Scheme of Studies/Examination w.e.f. 2023-24

Semester - III

S.No.	Category	Course Code	Course Title	Hours Per week			Total Contact Hrs. per week	Credits	Examination Schedule (Marks)			
				L	T	P			Marks of classwork	Theory	Practical	Total
1.	ESC		Digital Electronics	3	0	0	3	3	30	70		100
2.	PCC		Advanced-Data Structure	3		0	3	3	30	70		100
3.	PCC		Database Management Systems with SQL	3	0	0	3	3	30	70		100
4.	PCC		Programming with C++	3	0	0	3	3	30	70		100
5.	PCC		Introduction to AI and ML	3	0	0	3	3	30	70		100
6.	BSC		Calculus and Ordinary Differential Equations	3	0	0	3	3	30	70		100
7.	LC		Digital Electronics Lab	0	0	2	2	1	50		50	100
8.	LC		Advanced-Data Structure Lab	0	0	2	2	1	50		50	100
9.	LC		Database Management Systems Lab	0	0	2	2	1	50		50	100
10.	LC		Programming with C++ Lab	0	0	2	2	1	50		50	100
11.	MC		Constitution of India*	2	0	0	2	0	30	70		100*
			Total	20	0	8	28	22	380	420	200	1000

* The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree. However, these marks will be shown in the detailed marks certificate of the students.

DIGITAL ELECTRONICS

Semester	III				
Course code					
Category	Engineering Science courses				
Course title	Digital Electronics				
Scheme and Credits	L	T	P	Credits	
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVES:

1. To understand the basic theoretical concepts of digital systems like the binary system and Boolean algebra.
2. To use Boolean algebraic formulations to design digital systems. To design using combinational/sequential circuits.
3. To express real-life problems in logic design terminology.
4. To understand the logic of adders, subtractors and converters.

UNIT - I

FUNDAMENTALS OF DIGITAL SYSTEMS AND LOGIC FAMILIES

Digital signals, digital circuits, AND, OR, NOT, NAND, NOR and Exclusive-OR operations, Boolean algebra, examples of IC gates, number systems - binary, signed binary, octal hexadecimal number, binary arithmetic, one's and two's complements arithmetic, codes, error detecting and correcting codes.

UNIT - II

COMBINATIONAL DIGITAL CIRCUITS

Standard representation for logic functions, K-map representation, and simplification of logic functions using K-map, minimization of logical functions. Don't care conditions, Multiplexer, De-Multiplexer, Decoders, Adders, Subtractors, BCD arithmetic, carry look-ahead adder, serial adder, ALU, elementary ALU design, popular MSI chips, digital comparator, parity checker/generator, code converters, priority encoders, decoders/drivers for display devices, Q-M method of function realization.

UNIT - III

SEQUENTIAL CIRCUITS AND SYSTEMS

A 1-bit memory, the circuit properties of the Bistable latch, the clocked SR flip-flop, J-K flip-flop, T flip-flop and D flip-flop, applications of flip-flops, shift registers, applications of shift registers,

serial-to-parallel converter, parallel-to-serial converter, ring counter, sequence generator, ripple (Asynchronous) counters, synchronous counters, counters design using flip-flops, special counter IC's, asynchronous sequential counters, applications of counters.

UNIT - IV

A/D AND D/A CONVERTERS

Digital to analog converters: weighted resistor/converter, R-2-R Ladder D/A converter, specifications for D/A converters, examples of D/A converter ICs, sample and hold circuit, Analog to digital converters: quantization and encoding, parallel comparator A/D converter, successive approximation A/D converter, counting A/D converter, dual slope A/D converter.

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

- CO1: Outline the general concepts and terminology related to logic gates, logic families, combinational and sequential circuits.
- CO2: Discuss the basic analog/digital components and their interconnections in logic families and circuits.
- CO3: Apply different methods/techniques to design various digital circuits.
- CO4: Analyse day to day problems and industrial problems for their solutions using digital circuits.
- CO5: Contrast different types of digital circuits and their designing methods.
- CO6: Design digital circuit for various practical problems.

TEXT AND REFERENCE BOOKS:

1. R. P. Jain, "Modern Digital Electronics", McGraw Hill Education, 2009.
2. M. M. Mano, "Digital logic and Computer Design", Pearson Education India, 2016.
3. A. Kumar, "Fundamentals of Digital Circuits", Prentice Hall India, 2016.
4. Nasib Singh Gill and J B Dixit, "Digital Design and Computer Organization", University Science Press, New Delhi

ADVANCE DATA STRUCTURE

Semester	III				
Course code					
Category	Professional Core Courses				
Course title	Advance Data Structure				
Scheme and Credits	L	T	P	Credits	
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVES:

1. To understand and apply linear data structures-List, Stack and Queue.
2. To understand the tree algorithms and their applications.
3. To learn different algorithms and analysis techniques.
4. To apply sorting algorithms in real-time applications

UNIT - I

Review of Linear Data Structures

Linked List: Traverse, Insertion, Deletion; Circular List: Traverse, Insertion, Deletion, Doubly List and Circular List: Insertion and deletions; Stacks and Queue implementation using linked list.

Introduction to Dictionaries: Definition, Dictionary Abstract Data Type, Implementation of Dictionaries.

Hashing: Review of Hashing, Hash Function, Collision Resolution Techniques in Hashing, Separate Chaining, Open Addressing, Linear Probing, Quadratic Probing, Double Hashing, Rehashing, and Extendible Hashing.

UNIT - II

Advanced Trees: Trees: Review of binary trees and binary search trees: traversing, insertion, and deletion; **AVL Trees:** Introduction to AVL trees, Rotations in AVL trees: LL, RR, LR & RL,

Introduction, Search, Insert & delete operations: Red-Black Trees, 2-3 Trees, B-Trees, B+ Trees, Splay Trees.

UNIT - III

Sets: Representation on Sets, Operations on Sets, Application on Sets,

Files: File Concepts, File organization, Files and Streams, Working with Files Using I/O Stream, Sequential File Organization, Direct File Organization, Indexed Sequential Organization

UNIT - IV

Graphs: Representation, Basic terminology, traversal, connected components, shortest path, topological sort, Dijkstra's Algorithm, Floyd Warshall's Algorithm, network flow problems.

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

- CO1: Design and Analyze programming problem statements.
- CO2: Understand the ADT/libraries, and use it to design algorithms for a specific problem.
- CO3: Select algorithm design approaches in a problem-specific manner.
- CO4: Compare & contrast the complexity analysis of various sorting & searching algorithms.
- CO5: To be able to analyse the efficiency of algorithms.
- CO6: Implement various data structure concepts on real-world industrial problems.

TEXT AND REFERENCE BOOKS:

1. Seymour Lipschutz: Data Structures with C, Schaum's outline by TMH
2. E Horowitz and S. Sahni: Fundamentals of Data Structures in C, Second Edition, Universities Press, Hyderabad.
3. R.B. Patel: Expert Data Structures in C, Khanna Publishers, 2001.
4. R.L. Kruse: Data Structures & Program Design in C, PHI.
5. D.F. Knuth: The art of Computer Programming Vol 1, Narosa Publications, 1985.
6. Byron S. Gottfried & J K Chhabra: Theory and Problems of Programming with C Language, Schaum's Series, TMH, 2005.

DATABASE MANAGEMENT SYSTEMS WITH SQL

Semester	III				
Course code					
Category	Professional Core Courses				
Course title	Database Management Systems With SQL				
Scheme and Credits	L	T	P	Credits	
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVES:

1. To understand the different issues involved in the design and implementation of a database system.
2. To study the physical and logical database designs, database modeling, relational, hierarchical, and network models
3. To understand and use data manipulation language to query, update, and manage a database
4. To develop an understanding of essential DBMS concepts such as: database security, integrity, concurrency, distributed database, and intelligent database, Client/Server (Database Server), Data Warehousing.
5. To design and build a simple database system and demonstrate competence with the fundamental tasks involved with modeling, designing, and implementing a DBMS.

UNIT - I

Database system architecture: Data Abstraction, Data Independence, Data Definition Language (DDL), Data Manipulation Language (DML).

Data models: Entity-relationship model, network model, relational and object-oriented data models, integrity constraints, data manipulation operations.

UNIT - II

Relational query languages: Relational algebra, Tuple and domain relational calculus, SQL3, DDL and DML constructs, Open source and Commercial DBMS - MYSQL, ORACLE, DB2, SQL server.

Relational database design: Domain and data dependency, Armstrong's axioms, Normal forms, Dependency preservation, Lossless design.

Query processing and optimization: Evaluation of relational algebra expressions, Query equivalence, Join strategies, Query optimization algorithms.

UNIT - III

Transaction processing: Concurrency control, ACID property, Serializability of scheduling, Locking and timestamp-based schedulers, Multi-version and optimistic Concurrency Control schemes, Database recovery.

Storage strategies: Indices, B-trees, hashing,

UNIT - IV

Database Security: Authentication, Authorization and access control, DAC, MAC and RBAC models, Intrusion detection, SQL injection.

Advanced topics: Object-oriented and object-relational databases, Logical databases, Web databases, Distributed databases, Data warehousing and data mining.

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

- CO1: For a given query, write relational algebra expressions for that query and optimize the developed expressions
- CO2: For a given requirement specification, design the databases using E R method and normalization.
- CO3: For a given specification, construct the SQL queries for Open source and Commercial DBMS - MYSQL, ORACLE, and DB2.
- CO4: For a given query, optimize its execution using Query optimization algorithms
- CO5: For a given transaction-processing system, determine the transaction atomicity, consistency, isolation, and durability.
- CO6: Implement the isolation property, including locking, and time stamping based on concurrency control and Serializability of scheduling.

TEXT AND REFERENCE BOOKS:

1. Database System Concepts, 6th Edition by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill
2. Principles of Database and Knowledge–Base Systems, Vol 1 by J. D. Ullman, Computer Science Press.
3. Fundamentals of Database Systems, 5th Edition by R. Elmasri and S. Navathe, Pearson Education
4. Foundations of Databases, Reprint by Serge Abiteboul, Richard Hull, Victor Vianu, Addison-Wesley

PROGRAMMING WITH C++

Semester	III				
Course code					
Category	Professional Core Courses				
Course title	Programming with C++				
Scheme and Credits	L	T	P	Credits	
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVES:

1. To understand how the C++ is superset of C by incorporating the Object-oriented features in C language.
2. To learn how to efficiently use the memory using Pointers and Dynamic Memory Management
3. To learn how to implement different types of constructors and the use of destructor.
4. To learn how to implement the concept of data abstraction, encapsulation and how to perform different types of overloading i.e. operators and functions.
5. To learn how inheritance helps to reuse the code.
6. To learn how we can implement dynamic binding with polymorphism.
7. To learn the use of exception handling in C++ programs.

UNIT - I

Object-Oriented Programming Concepts: Introduction, comparison between procedural programming paradigm and object-oriented programming paradigm, basic concepts of object-oriented programming — concepts of an object and a class, interface and implementation of a class, operations on objects, relationship among objects, abstraction, encapsulation, data hiding, inheritance, overloading, polymorphism, messaging.

Classes and Objects: Specifying a class, creating class objects, accessing class members, access specifiers, static members, use of const keyword, friends of a class, empty classes, nested classes, local classes, abstract classes, container classes, bit fields and classes.

UNIT - II

Inheritance: Introduction, defining derived classes, forms of inheritance, ambiguity in multiple and multipath inheritances, virtual base class, object slicing, overriding member functions, object composition and delegation, order of execution of constructors and destructors.

Pointers and Dynamic Memory Management: Declaring and initializing pointers, accessing data through pointers, pointer arithmetic, memory allocation (static and dynamic), dynamic memory management using new and delete operators, the pointer to an object, this pointer, pointer

related problems - dangling/wild pointers, null pointer assignment, memory leak and allocation failures.

UNIT - III

Constructors and Destructors: Need for constructors and destructors, copy constructor, dynamic constructors, explicit constructors, destructors, constructors and destructors with static members, initializer lists.

Operator Overloading and Type Conversion: Overloading operators, rules for overloading operators, overloading of various operators, type conversion - basic type to class type, class type to basic type, class type to another class type.

Virtual functions & Polymorphism: Concept of binding - early binding and late binding, virtual functions, pure virtual functions, abstract classes, virtual destructors.

UNIT - IV

Exception Handling: Review of traditional error handling, basics of exception handling, exception handling mechanism, throwing mechanism, catching mechanism, rethrowing an exception, specifying exceptions.

Templates and Generic Programming: Template concepts, Function templates, class templates, illustrative examples.

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

CO1: Understand the concept of Object-Oriented Programming through C++.

CO2: Identify importance of object-oriented programming and difference between Procedural programming and object oriented programming features.

CO3: be able to make use of objects and classes for developing programs.

CO4: be able to use various object-oriented concepts to solve different problems.

CO5: be able to develop the programs /Projects using some advanced features of C++ Programming.

TEXT AND REFERENCE BOOKS:

1. Bjarne Stroustrup, "C++ Programming language", 3rd edition, Pearson education Asia (1997)
2. Lafore R. "Object oriented Programming in C++", 4th Ed. Techmedia, New Delhi (2002).
3. Yashwant Kenetkar, "Let us C++", 1st Ed., Oxford University Press (2006)
4. B.A. Forouzan and R.F. Gilberg, Compiler Science, "A structured approach using C++" Cengage Learning, New Delhi.

CALCULUS AND ORDINARY DIFFERENTIAL EQUATIONS

Semester	III				
Course code					
Category	Basic Science Courses				
Course title	Calculus and Ordinary Differential Equations				
Scheme and Credits	L	T	P	Credits	
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVES:

1. Understand the basic concepts of Limits & Continuity.
2. Understand partial and total derivative and apply it to real time applications.
3. Understand the ordinary Differential Equations of First, Second and Higher Order.

UNIT - I

Multivariable Differential Calculus: Limit, Continuity and Partial derivatives, Homogeneous functions, Euler's Theorem, Total derivative, Maxima, Minima and Saddle points, Lagrange's method of undetermined multipliers.

UNIT - II

Multivariable Integral Calculus: Double integral, change of order of integration, Change of variables, Applications of double integral to find area enclosed by plane curves, Triple integral.

UNIT - III

Ordinary Differential Equations of first order: Linear and Bernoulli's equations, Exact differential equations, Equations reducible to exact differential equations, Applications of differential equations of first order and first degree to simple electric circuits, Newton's law of cooling, Heat flow and Orthogonal trajectories.

UNIT - IV

Ordinary Differential equations of second and higher order: Linear differential equations of second and higher order, Complete solution, Complementary function and Particular integral, Method of variation of parameters to find particular integral, Cauchy's and Legendre's linear equations, Simultaneous linear differential equations with constant coefficients, Applications of linear differential equations to oscillatory electric circuits.

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

CO1: Deal with functions of several variables and evaluate partial derivative.

CO2: Evaluate multiple integrals and their usage.

CO3: Solve ordinary differential equations that model physical processes.

CO4: Formulate and solve problems involving moment of inertia, volume and centre of gravity.

CO5: Solve engineering problems related to oscillatory electric circuits.

CO6: Solve field problems in engineering involving Ordinary Differential Equations like R-L-C circuits and to find heat loss

TEXT AND REFERENCE BOOKS:

1. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, Pearson Education.
2. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons.
3. B. V. Ramana, Higher Engineering Mathematics, Tata McGraw-Hill Publishing Company Limited.
4. N. P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications.
5. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers.
6. W. E. Boyce and R. C. DiPrima, Elementary Differential Equations and Boundary Value Problems, Wiley India.
7. S. L. Ross, Differential Equations, Wiley India.
8. E. A. Coddington, An Introduction to Ordinary Differential Equations, Prentice Hall India.
9. E. L. Ince, Ordinary Differential Equations, Dover Publications

INTRODUCTION TO AI and ML

Semester	III				
Course code					
Category	Professional Core Courses				
Course title	Introduction to AI and ML				
Scheme and Credits	L	T	P	Credits	
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVES:

1. Understand the very basics and Uses of Artificial Intelligence (AI)
2. Understand the basics and uses of Machine Learning (ML)
3. To provide the most fundamental knowledge to the students so that they become familiar with basic principles of AI towards problem solving, inference, knowledge representation and learning
4. Understand the logic-building methods and inferences for the knowledge representation.
5. Explore application of AI techniques in Expert systems, Neural Networks.

UNIT - I

Introduction to AI: What is AI, Turing test, History of AI, Artificial Intelligence Techniques, advantages, and limitations of AI, Impact and Examples of AI

Applications of AI by domain: Transportation, home/service robots, healthcare, education, low-resource communities, public safety and security, employment and workplace, entertainment, finance, banking and insurance.

Introduction to Machine Learning: What is Machine Learning, History of Machine Learning, Machine Learning and Statistics, Types of Machine Learning – Supervised, Unsupervised, Semi-supervised, Reinforcement Learning,

UNIT - II

Intelligent agent: Nature of Agents, Rationality and Rational agent with performance measures Flexibility and Intelligent agents, Task environment and its properties, Types of agents, other aspects of agents;

Multi-Agent Systems: Agents and Objects; Agents and Expert Systems; Generic Structure of Multiagent System, Semantic Web, Agent Communication, Knowledge Sharing using Ontologies, Agent Development Tools.

UNIT - III

Knowledge Representation schemes and reasoning: Approaches and issues, procedural vs declarative knowledge, Matching, conflict resolution.

Logic: Propositional logic, predicate logic, Resolution, Resolution in propositional logic and predicate logic, Clause form, unification algorithm.

Uncertain Knowledge and reasoning: Methods, Bayesian probability and belief network, Probabilistic reasoning, Forward and backward reasoning, Other uncertain techniques-Data mining, Fuzzy logic, Dempster -Shafer theory.

UNIT - IV

Planning: The Planning problem, planning with state space search, partial order planning, planning graphs, planning with propositional logic, Analysis of planning approaches, Hierarchical planning, conditional planning, Continuous and Multi Agent planning.

Learning: Introduction to Learning, Types of Learning: Learning by Induction, Rote Learning, Symbol Based Learning, Identification Trees, Explanation Based Learning, Transformational Analogy, Introduction to Neural Networks, Expert Systems.

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

CO1: Formulate a problem and build intelligent agents.

CO2: Apply basic principles of AI in solutions that require problem solving, inference, knowledge representation and learning.

CO3: Analyze the problem and infer new knowledge using suitable knowledge representation schemes.

CO4: Develop planning and apply learning algorithms on real world problems.

CO5: Design an expert system and implement advance techniques in Artificial Intelligence.

CO6: Create a real life and industrial problems related mini project.

TEXT AND REFERENCE BOOKS:

1. Artificial Intelligence 3e: A Modern Approach Paperback – By Stuart J Russell & Peter Norvig; Publisher – Pearson
2. Artificial Intelligence Third Edition by Kevin Knight, Elaine Rich, B. Nair – Mc Graw Hill
3. Artificial Intelligence Third Edition by Patrick Henry Winston – Addison-Wesley Publishing Company
4. Machine Learning using Pythons, U Dinesh Kumar, Manaranjan Pradhan, John Wiley & Sons.
5. A Classical Approach to Artificial Intelligence, M. C. Trivedi, Khanna Publishing House.
6. Machine Learning, V. K. Jain, Khanna Publishing House.
7. Advanced Data Analytics Using Python: With Machine Learning, Deep Learning, Sayan Mukhopadhyay, Apress.
8. Introduction to Machine Learning, Jeeva Jose, Khanna Publishing House.

CONSTITUTION OF INDIA

Semester	III				
Course code					
Category	Mandatory courses				
Course title	Constitution of India				
Scheme and Credits	L	T	P	Credits	
	2	0	0	-	
Classwork	30				
Exam	70				
Total	100				
Duration of Exam	3 hrs				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVES:

1. Understand the premises informing the twin themes of liberty and freedom from a civil rights perspective.
2. To address the growth of Indian opinion regarding modern Indian intellectuals' constitutional role and entitlement to civil and economic rights as well as the emergence of nationhood in the early years of Indian nationalism.
3. To address the role of socialism in India after the commencement of the Bolshevik Revolution in 1917 and its impact on the initial drafting of the Indian Constitution.
4. To learn procedure and effects of emergency, composition and activities of election commission and amendment procedure.

UNIT - I

Philosophy of Indian Constitution: Salient features of Indian Constitution, Preamble, and Nature of Indian Constitution, Procedure for amendment of the Constitution.

UNIT - II

Federal structure and distribution of legislative and financial powers between the Union and the States

UNIT - III

Organs of Governance: President – Qualification and Powers of the President, Governor- Qualification and Powers of Governor,

Parliament: Composition, Qualifications and Disqualifications, Judiciary: Appointment, Tenure and Removal of Judges.

UNIT - IV

Fundamental Rights: Origin and development of Fundamental rights, Need for fundamental rights. Introduction to Rights to equality, right to freedom, right against exploitation, Right to

freedom of religion, Cultural and Education rights and Fundamental duties.

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

- CO1: Discuss the growth of the demand for civil rights in India for the bulk of Indians before the arrival of Gandhi in Indian politics.
- CO2: Discuss the intellectual origins of the framework of argument that informed the conceptualization of social reforms leading to a revolution in India.
- CO3: Exercise his fundamental rights in proper sense at the same time identifies his responsibilities in national building.
- CO4: Discuss the circumstances surrounding the foundation of the Congress Socialist Party [CSP] under the leadership of Jawaharlal Nehru and the eventual failure of the proposal of direct elections through adult suffrage in the Indian Constitution.
- CO5: Discuss the passage of the Hindu Code Bill of 1956.
- CO6: Analyse the Indian political system, the powers and functions of the Union, State and Local Governments in detail.

TEXT AND REFERENCE BOOKS:

1. The Constitution of India, 1950 (Bare Act), Government Publication.
2. Dr. S.N. Busi, Dr. B.R. Ambedkar framing of Indian Constitution, latest Edition
3. M.P. Jain, Indian Constitution Law, Lexis Nexis, latest edition
4. D.D. Basu, Introduction to Constitution of India, Lexis Nexis, latest edition.

NOTE: The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree. However, these marks will be shown in the detailed marks certificate of the students.

DIGITAL ELECTRONICS LAB

Semester	III				
Course code					
Category	Laboratory course				
Course title	Digital Electronics Lab				
Scheme and Credits	L	T	P	Credits	
	0	0	2	1	
Classwork	50 Marks				
Exam	50 Marks				
Total	100 Marks				
Duration of Exam	02 Hours				

Note:

- (i) At least 10 experiments are to be performed by students in the semester.
- (ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus

CONTENTS

Implementation of all experiments with the help of Bread-Board.

1. Study of Logic Gates: Truth-table verification of OR, AND, NOT, XOR, NAND and NOR gates; Realization of OR, AND, NOT and XOR functions using universal gates.
2. Half Adder / Full Adder: Realization using basic and XOR gates.
3. Half Subtractor / Full Subtractor: Realization using NAND gates.
4. 4-Bit Binary-to-Gray & Gray-to-Binary Code Converter: Realization using XOR gates.
5. 4-Bit and 8-Bit Comparator: Implementation using IC7485 magnitude comparator chips.
6. Multiplexer: Truth-table verification and realization of Half adder and Full adder.
7. Demultiplexer: Truth-table verification and realization of Half subtractor and Full subtractor.
8. Flip Flops: Truth-table verification of JK Master Slave FF, T-type and D-type FF.
9. Asynchronous Counter: Realization of 4-bit up counter and Mod-N counter.
10. Synchronous Counter: Realization of 4-bit up/down counter and Mod-N counter.
11. Shift Register: Study of shift right, SIPO, SISO, PIPO, PISO & Shift left operations.
12. DAC Operation: Study of 8-bit DAC, obtain staircase waveform.
13. ADC Operations: Study of 8-bit ADC

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

- CO1: Define different types of logic gates, identify their ICs and also verify their truth table.
- CO2: Derive basic logic gates, adder, and subtractor using universal gates.
- CO3: Illustrate realization of Boolean expression in SOP and POS form and design it using logic gates.
- CO4: Design and test combinational circuits.
- CO5: Design and develop sequential circuits.
- CO6: Demonstrate team- based laboratory activities with fellow students to interact effectively on a social and interpersonal level.

DATABASE MANAGEMENT SYSTEM LAB

Semester	III				
Course code					
Category	Laboratory course				
Course title	Database Management System Lab				
Scheme and Credits	L	T	P	Credits	
	0	0	2	1	
Classwork	50 Marks				
Exam	50 Marks				
Total	100 Marks				
Duration of Exam	02 Hours				

Note:

- (i) At least 10 experiments are to be performed by students in the semester.
- (ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus

CONTENTS

1. Design a Database and create required tables. For e.g. Bank, College Database
2. Apply the constraints like Primary Key, Foreign key, NOT NULL to the tables.
3. Write a SQL statement for implementing ALTER, UPDATE and DELETE.
4. Write the queries to implement the joins.
5. Write the query for implementing the following functions: MAX (), MIN (), AVG () and COUNT ().
6. Write the query to implement the concept of Integrity constraints.
7. Write the query to create the views.
8. Perform the queries for triggers.
9. Study of PL/SQL block.
10. Write a PL/SQL block to satisfy some conditions by accepting input from the user.
11. Write the query for creating the users and their role. Mini project (Application Development using Oracle/ MySQL)
 - a. Inventory Control System
 - b. Material Requirement Processing.
 - c. Hospital Management System.
 - d. Railway Reservation System.
 - e. Personal Information System.
 - f. Web-Based User Identification System.
 - g. Time Table Management System.
 - h. Hotel Management

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

- CO1: Identify the fundamental elements of relational database management systems.
- CO2: Design and explain the basic concepts of relational data model, entity-relationship model, and relational database design.
- CO3: Apply the relational database theory to formulate basic and advanced SQL queries and relational algebra expressions for the queries.
- CO4: Identify the use of normalization and functional dependency in database design.

CO5: Understand the concept of transactions and serializability in database management system.

CO6: Classify the implementation details of Concurrency control protocols and discuss various database recovery methods.

ADVANCED DATA STRUCTURE LAB

Semester	III					
Course code						
Category	Laboratory course					
Course title	Advance Data Structure Lab					
Scheme and Credits	L	T	P	Credits		
	0	0	2	1		
Classwork	50 Marks					
Exam	50 Marks					
Total	100 Marks					
Duration of Exam	02 Hours					

Note:

- (i) At least 10 experiments are to be performed by students in the semester.
- (ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus

CONTENTS

1. Write a program to implement all operations on 1-D array.
2. Write a program to implement all operations on Simple Linked List.
3. Write a program to implement all operations on a circular Linked List.
4. Write a program to implement all operations on a doubly Linked List.
5. Write a program to implement all operations on a doubly circular Linked List.
6. Write a program to implement all operations on Stack using Array.
7. Write a program to implement all operations on Stack using Linked List.
8. Write a program to implement all operations on Queue using Array.
9. Write a program to implement all operations on Queue using Linked List.
10. Write a Program to implement dictionary techniques.
11. Write a program to implement hashing techniques.
12. Write a Program to implement Red-Black Trees.
13. Write a Program to implement Binary Search Trees.
14. Write a Program to design a menu to implement: Quick, Merge, and Bubble sorting.
15. Write a Program to develop a recursive Program to implement Breadth First Search and Depth First Search.
16. Write a Program to develop a non-recursive Program to implement Breadth First Search and Depth First Search

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

CO1: Identify the appropriate data structure for a given problem.

CO2: Implement Dictionary by using hashing techniques.

CO3: Analyse various basic operations of trees to improve the efficiency.

CO4: Build a Binary Heap using Priority queues.

CO5: Apply the concepts of data structures in various real-world applications.

CO6: Identify, model, solve and develop algorithms for real-life problems like shortest path and MST using graph theory.

PROGRAMMING WITH C++ LAB

Semester	III				
Course code					
Category	Laboratory course				
Course title	Programming with C++ Lab				
Scheme and Credits	L	T	P	Credits	
	0	0	2	1	
Classwork	50 Marks				
Exam	50 Marks				
Total	100 Marks				
Duration of Exam	02 Hours				

Note:

- (i) At least 10 experiments are to be performed by students in the semester.
- (ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus

CONTENTS

1. Write a program that uses a class where the member functions are defined inside a class.
2. Write a program that uses a class where the member functions are defined outside a class.
3. Write a program to demonstrate the use of static data members.
4. Write a program to demonstrate the use of const data members.
5. Write a program to demonstrate the use of zero argument and parameterized constructors.
6. Write a program to demonstrate the use of dynamic constructor.
7. Write a program to demonstrate the use of explicit constructor.
8. Write a program to demonstrate the use of initializer list.
9. Write a program to demonstrate the overloading of increment and decrement operators.
10. Write a program to demonstrate the overloading of binary arithmetic operators.
11. Write a program to demonstrate the overloading of memory management operators.
12. Write a program to demonstrate the multilevel inheritance.
13. Write a program to demonstrate the multiple inheritance.
14. Write a program to demonstrate the virtual derivation of a class.
15. Write a program to demonstrate the runtime polymorphism.
16. Write a program to demonstrate the exception handling.
17. Write a program to demonstrate the use of function template.
18. Write a program to demonstrate the use of class template.

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

CO1: Understand dynamic memory management techniques using pointers, constructors, destructors, etc.

CO2: Describe the concept of function overloading, operator overloading, virtual functions and polymorphism.

CO3: Classify inheritance with the understanding of early and late binding

CO4: Usage of exception handling and generic programming.

CO5: Develop the programs /Projects using some advanced features of C++ Programming.

CO6: Percept the utility and applicability of OOP.